

LAU KAI KWAN

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EDUCATION

The Chinese University of Hong Kong

Aug 2026 - Present

- M.Phil in Mathematics

The Chinese University of Hong Kong (CUHK)

Aug 2022 - Jul 2026

- B.Sc in Mathematics and minor in Physics
- Selected for an exclusive interview in the [Department of Mathematics Newsletter \(Issue 22\)](#)

Ng Yuk Secondary School

Sep 2015 - Jun 2021

- Sir Edward Youde Memorial Prize
 - HKDSE: 29/35 for best five score subjects
 - SAT: 800/800 Math II, 750/800 Math I
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PUBLICATIONS, PREPRINTS & SERVICE

- [PhyloSDF: Phylogenetically-Conditioned Neural Generation of 3D Skull Morphology via Residual Flow Matching](#), **Kaikwan Lau** and Gary P. T. Choi, ArXiv Preprint.
 - [Are Vocaloid Songs Getting Denser? A Longitudinal Audio Analysis of 1,900 Hatsune Miku Songs \(2007--2025\)](#), **Kaikwan Lau**, Accepted to the 146th Information Processing Society of Japan Special Interest Group on Music and Computer Information.
 - [Geometric and Statistical Analysis of Avian Skull Morphology](#), **Kaikwan Lau** and Gary P. T. Choi, ArXiv Preprint.
 - [Efficient Score Pre-computation for Diffusion Models via Cross-Matrix Krylov Projection](#), **Kaikwan Lau**, Andrew S. Na, and Justin W. L. Wan, Pattern Recognition (ICPR 2026), Lecture Notes in Computer Science (vol. 16813), pp. 697-712, 2026.
 - Service: Reviewer, Tackling Climate Change with Machine Learning Workshop, NeurIPS 2026
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PROFESSIONAL & RESEARCH EXPERIENCE

NVIDIA AI Technology Center (NVAITC), Student Researcher

Mar 2025 - Jul 2026

- Conducted advanced AI research under the mentorship of Ivan Au Yeung and Juntao about down scaling problem for weather prediction.
- Prepared and two workshop papers targeted for publication in NeurIPS 2026.

The Chinese University of Hong Kong, Research Assistant

Mar 2025 - Jul 2026

- Developed custom MATLAB and Python algorithms to perform high-dimensional geometric and statistical analysis on biological structures.
- Quantified complex avian skull morphology under the supervision of Prof. Gary P. T. Choi.

University of Wollongong College Hong Kong, Lecturer

Jan 2026 - May 2026

- Facilitate specialized tutorials and lectures on applied mathematics and probability theory for the Diploma of Applied Education.
- Deliver complex mathematical concepts to diverse student groups, ensuring high engagement and comprehension.

Quality Education Fund Project Funded by the Education Bureau, Programmer and Designer

Jan 2024 - Dec 2025

- Collaborated with EdTech company OneEd to secure HKD 250 million in funding for educational innovation.
- Architected "Lambda Math," a comprehensive digital learning ecosystem serving secondary students across Hong Kong.
- Developed AI-driven generative models to produce dynamic, adaptive examination questions.

The University of Waterloo, Visiting Scholar

Jun 2025 - Aug 2025

- Conducted research on numerical methods for Generative AI under the supervision of Prof. Justin Wan.
- Optimized Generative AI inference by reformulating score computation through Partial Differential Equations (PDEs) via PyTorch, achieving 100x faster generation speeds compared to standard DDPM.
- Engineered a novel Cross-Matrix Krylov Projection algorithm, successfully reducing computational overhead by 40%.

TEACHING & MENTORSHIP EXPERIENCE

- The Chinese University of Hong Kong, Research Assistant** **Jan 2025 - Mar 2025**
- Supervised by Dr. WONG Chak Fu Jeff in Department of Mathematics and Information Technology.
 - Summarized and checked research papers about Game Theory.
- The Education University of Hong Kong, Research Assistant** **Aug 2024 - Oct 2024**
- Supervised by Prof. SUEN Chun Kit Anthony of Department of Mathematics and Information Technology.
 - Summarized and analyzed research papers about Partial Differential Equations using LaTeX.
 - Designed master-level mathematics course assignments.
- First-Year Mathematics Honours Scheme (CUHK), Teaching Assistant** **Sep 2023 - May 2025**
- Conducted and designed university-mathematics notes weekly for Year 1 Mathematics major students
- Training Programme for Young Mathematics Talents (CUHK), Teaching Assistant** **Jul 2024 - Aug 2024**
- Instructed advanced courses in Linear Algebra I and Game Theory for high-achieving high school students. in the Greater Bay Area.
 - Mentored 4 students in independent mathematical research on Game Theory; successfully guided the team to achieve awards for "Best Research Topic" and "Best Presentation."
 - Cooperated with all teaching assistants to hold final exams and tutorials.
- Enrichment Programme for Young Mathematics Talents (CUHK), Teaching Assistant** **Jul 2024 - Aug 2024**
- Delivered instructional materials and assessments in English for the Differential Geometry course.
 - Mentored young talents by fostering their mathematical maturity to prepare them for university-level courses.
- University of Wollongong College Hong Kong, Teaching Assistant** **Dec 2024 - Dec 2025**
- Conducted tutorial classes and lectures for students about applied mathematics in probability theory in Diploma of Applied Education.
- Program for the Gifted and Talented (CUHK), Teaching Assistant** **Dec 2023**
- Communicated to gifted and talented students effectively.
 - Maintained a disciplined classroom environment.
- Ng Yuk Secondary School, Mathematics Tutor** **Jan 2022 - Present**
- Raised at least one DSE exam levels in core maths subject for low banding students.
 - Organized and led study groups.

LEADERSHIP, HONORS, & AWARDS

- Chung Chi Astronomy Club (CUHK), President** **Apr 2024 - March 2026**
- Oversaw all administrative affairs and college-level communications to ensure institutional compliance and operational excellence.
 - Managed an 8-member executive committee to deliver high-impact programming organization.
- CUHK Astronomy Club, Publicity Secretary** **Mar 2023 - Feb 2024**
- Published astronomy-related posts on social media and produced a souvenir for the public at the art fair.
 - Designed website and 24-page reports for a 400-member student body.
 - Organized an astronomy open contest with participation from 20 teams.

SCHOLARSHIPS

- Hsin Chong-K.N. Godfrey Yeh Education Fund Bursaries 2024/26
- Faculty Exchange Scholarship Scheme 2026
- Centaline Charity Fund Scholarship 2025-2026
- Yasumoto International Exchange Scholarships
- 1978 Mathematics Alumnus Li Sze-lim Scholarship 2025/26
- Spring Dinner Scholarship 2025
- Hsin Chong-K.N. Godfrey Yeh Education Fund Bursaries 2024/25
- Centaline Charity Fund Scholarship 2024-2025
- ~Angel~ Care Student Assistance Project
- Centaline Charity Fund Scholarship 2023-2024
- Mathematics scholarship by Department of Mathematics, CUHK
- Chung Chi College admission scholarship for outstanding extra-curricular performance
- Faculty admission scholarship by Faculty of Science, CUHK
- Centaline Charity Fund Scholarship 2022-2023
- Sir Edward Youde Memorial Prize 2020-2021

LEADERSHIP, HONORS, & AWARDS

Hong Kong and Macau Math Olympiad Open Contest

- Bronze Award (2015-2017)
- Silver Award (2018-2019)

China National Math Olympiad - Hua Xia Cup

- Third Class Award (2017-2018)

Hong Kong International Youth Performance Arts Festival

- Champion in Flute
- First-runner up in Flute